

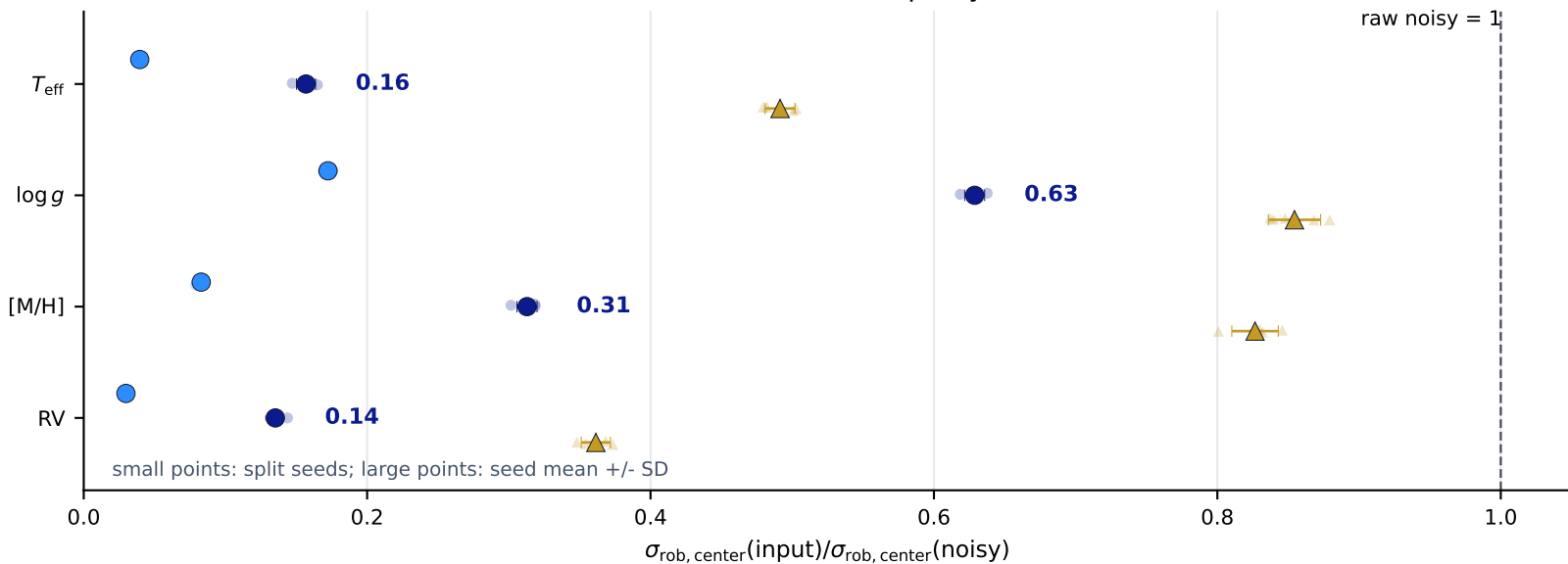
A. Clean-domain transfer protocol



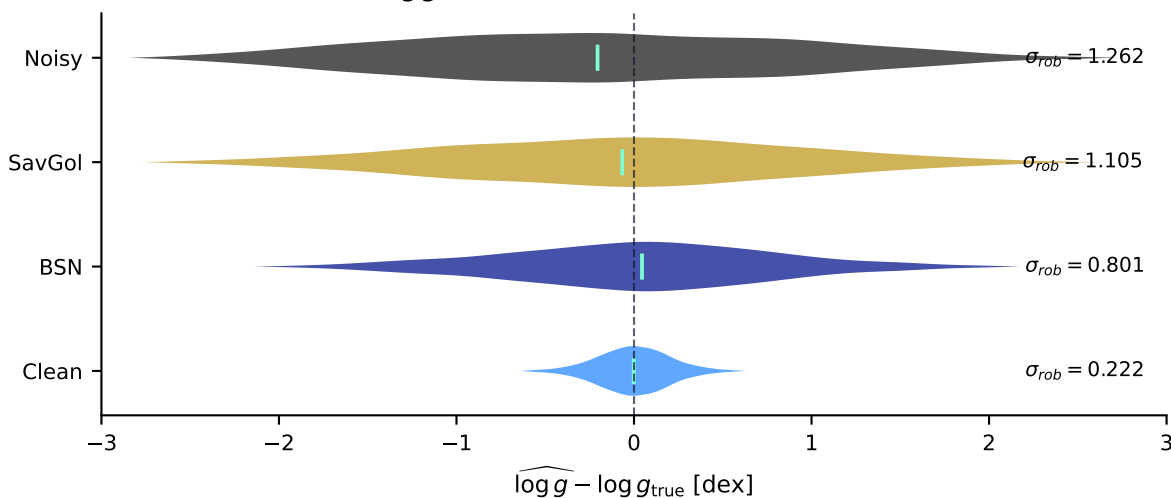
trained only on clean spectra; frozen before noisy, smoothed, or BSN-denoised evaluation

● Clean ● BSN ▲ SavGol

B. Frozen clean-trained proxy errors



C. $\log g$ residuals for one frozen calibrator



$\log g$ fixed split
 σ_{rob} : noisy 1.262 dex
 BSN 0.801 dex
 SavGol 1.105 dex
 clean reference 0.222 dex

BSN/noisy:
 σ_{rob} ratio 0.636 [0.609, 0.662]
 MAE ratio 0.700 [0.683, 0.717]
 R^2 : noisy 0.011, BSN 0.463, clean 0.940